

Unemployment Analysis with Python

CodeAlpha Data Science Internship — Task 2

1. Introduction

This project analyzes unemployment-rate data in India using Python. The analysis focuses on data cleaning, exploratory analysis, visualization of unemployment trends, comparison of unemployment before and during the COVID-19 period, and monthly pattern analysis.

2. Objectives

- Clean and prepare the unemployment dataset for analysis.
- Explore unemployment-rate data using descriptive statistics.
- Visualize unemployment trends over time.
- Investigate changes in unemployment during the COVID-19 period.
- Identify monthly patterns and possible seasonal trends.
- Present data-based findings relevant to economic and social discussions.

3. Dataset Description

The dataset contains unemployment-related observations for regions in India. The main variables include Region, Date, Frequency, Estimated Unemployment Rate (%), Estimated Employed, Estimated Labour Participation Rate (%), and Area.

4. Data Cleaning and Preparation

The original dataset contained 768 rows and 7 columns. During cleaning, 28 completely empty rows were removed, leaving 740 valid records. Column-name whitespace was removed using Pandas, and the Date column was converted to a datetime format. After cleaning, all seven columns had zero missing values.

Item	Result
Initial rows	768
Columns	7
Completely empty rows removed	28
Final rows	740
Missing values after cleaning	0

5. Exploratory Data Analysis

The unemployment-rate column was analyzed using descriptive statistics and visualized over time. The time-series graph showed substantial variation across the observations. Because the dataset contains multiple regions and Rural/Urban observations, the combined trend graph contains overlapping observations.

6. COVID-19 Impact Analysis

Period	Average Unemployment Rate
Before COVID (before March 2020)	9.51%
COVID period (March–December 2020)	17.77%
Difference	8.26 percentage points

The average unemployment rate increased from approximately 9.51% before March 2020 to 17.77% during March–December 2020, a difference of approximately 8.26 percentage points. This is a substantial increase in the period represented by the dataset. The result is a dataset-based association and does not by itself establish that COVID-19 was the only cause of the increase.

7. Monthly Pattern Analysis

Month	Average Rate
January	9.95%
February	9.96%
March	10.70%
April	23.64%
May	16.65%
June	10.55%
July	9.03%
August	9.64%
September	9.05%
October	9.90%

November	9.87%
December	9.50%

April recorded the highest monthly average unemployment rate at 23.64%, followed by May at 16.65%. Most other months were around 9–11%. Because the April and May peaks overlap with the COVID-19 period represented in the dataset, these peaks should not automatically be interpreted as a purely recurring seasonal effect.

8. Key Findings

- After cleaning, the dataset contained 740 valid records and 7 columns.
- No missing values remained after removing completely empty rows.
- The average unemployment rate was 9.51% before March 2020 and 17.77% during March–December 2020.
- The difference between the two COVID comparison periods was approximately 8.26 percentage points.
- April had the highest monthly average unemployment rate at 23.64%, followed by May at 16.65%.
- July had the lowest monthly average among the calculated monthly averages at 9.03%.

9. Tools and Technologies

Python, Pandas, and Matplotlib were used for data loading, cleaning, analysis, and visualization.

10. Conclusion

This project demonstrated how Python can be used to analyze unemployment data in India. The analysis included data cleaning, descriptive exploration, time-based visualization, COVID-period comparison, and monthly pattern analysis. The dataset showed a substantial increase in average unemployment during the COVID-19 period, while April and May showed particularly high monthly averages. These findings provide a data-based view of unemployment changes that can support further economic and social analysis.